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Youzhi Wang

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Education

Columbia University
MS Computer Science
GPA: 3.96

New York, NY
May 2024

relevant courses: Data Structures and Algorithms , DataBases

University of California, Santa Barbara
BS (double major) Applied Mathematics, Statistics and Data science
relevant courses: Cryptography, Automata and Formal Languages
Graduated with Honors and Major(Applied Math) distinction

Santa Barbara, CA
June 2023

Projects

NATURAL LANGUAGE PROCESSING PROJECT

Creating robust datasets for LLM evaluation (New York, NY) **May 2024 – August 2024**

- Coauthored a paper for EMNLP 2024 conference (top conference in NLP)
- Wrote Python scripts to generate 400 math questions and answers for evaluating LLMs
- Parameterized math question using Python to introduce randomness to better evaluate LLM's reasoning ability

DEEP LEARNING PROJECTS

Schizophrenia classification (New York, NY) **Jan 2024 – May 2024**

- Wrote Python scripts to train Neural Networks on 3D MRI brain images that achieved AROC of 0.87
- Wrote Python scripts to visualize regions of the brain that cause schizophrenia
- Wrote Python scripts that compared the performance of 2D vision transformers and 3D Neural networks

Cancer cell detection and classification (New York, NY) **Jan 2024 – May 2024**

- Wrote Python scripts to fine tuned vision transformer model to detect position of cancer cell in a image
- Ran Python scripts on Amazon Web Service servers to train Neural Network models

Multiple Sclerosis classification using Brain MRI (New York, NY) **Sep 2023 – Jan 2024**

- Wrote Python scripts using PyRadiomics to run radiomics analysis for 100 brain MRIs using clusters
- Wrote Bash scripts that runs brain MRI segmentation using FreeSurfer
- Wrote Python scripts using PyTorch and MONAI for classification tasks
- Ran Python and Bash scripts on a cluster computer through SSH and Sun Grid Engine

Earthquake data collection and classification (Santa Barbara, CA) **Jul 2022 – Jun 2023**

- Received Applied math Major Distinction award from the University of California, Santa Barbara for my work
- Collected and cleaned large collections of seismic signals (over 100,000 waveforms) using Python and ObsPy
- Wrote Python code in JupyterNotebooks that improved high-quality signal identification time by 5 times through restructuring the previous data processing pipeline
- Wrote Python code in JupyterNotebooks to label 200+ high-quality Earthquake signals that can be used to train ML algorithms
- Wrote Python code to test 5 different Machine Learning algorithms (KNN, LDA/QDA, SVMs and DCNN) for classifying high quality seismic wave signals

OTHER CODING PROJECTS

HTTP 1.0 Web server **Sep 2023 – Dec 2023**

- Wrote a HTTP 1.0 web server using C and socket APIs with image hosting and an interactive search module that uses a linked list to search through a given database.

Command Line Anagram Finder **Sep 2023 – Dec 2023**

- Wrote a command line Java application to find all the anagrams of a given word from a large text document using Binary Search Trees, AVL Trees and HashMaps

SKILLS

Proficient with: Python, PyTorch, sklearn, Java, Bash, SQL, AWS, Linux, Git, C/C++, Vim/Neovim, Tmux
Proficient languages: Mandarin